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Temperature coefficient of offset compensation for resistance bridge

Abstract

Systems and methods for temperature coefficient of offset compensation for a resistance bridge are disclosed. In one aspect, one or more current sources are added in parallel to resistance elements within a resistance bridge. The current source(s) may be selectively switched on and adjusted by a control circuit based on readings from a temperature sensor. In this fashion, the temperature induced variations in the resistance may be canceled or corrected allowing for better performance of the resistance bridge.

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Background/Summary

PRIORITY APPLICATION

(1) The present application is related to U.S. Patent Provisional Application Ser. No. 63/368,556

filed on Jul. 15, 2022, and entitled “TEMPERATURE COEFFICIENT OF OFFSET COMPENSATION FOR RESISTANCE BRIDGE,” the contents of which is incorporated herein by reference in its entirety.

BACKGROUND

I. Field of the Disclosure

(2) The technology of the disclosure relates generally to pressure sensing circuits such as a piezo resistor bridge and, more particularly, to a circuit that provides temperature compensation for the same.

II. Background

(3) Computing devices abound in modern society. The prevalence of these computing devices is driven in part by the many functions that are now enabled on such devices. With the advent of the myriad functions available to such devices, there has been an increase in processing performed by the devices, which, in turn generates heat and causes temperature changes in the circuits that provide these functions. While some circuits are not sensitive to temperature changes, other circuits such as piezoelectric-based pressure sensors are sensitive to temperature changes. One such circuit is a piezo resistor bridge such as may be used in a micro-electrical mechanical system (MEMS) sensor. Such sensitivity provides room for innovation.

SUMMARY

(4) Aspects disclosed in the detailed description include systems and methods for temperature coefficient of offset compensation for a resistance bridge. In particular, exemplary aspects of the present disclosure contemplate adding one or more current sources in parallel to resistance elements within a resistance bridge. The current source(s) may be selectively switched on and adjusted by a control circuit based on readings from a temperature sensor. In this fashion, the temperature-induced variations in the resistance may be canceled or corrected allowing for better performance of the resistance bridge.

(5) In this regard in one aspect, a pressure sensor is disclosed. The pressure sensor comprises a resistance bridge. The resistance bridge comprises a first resistor positioned between a first terminal node and a second terminal node. The resistance bridge also comprises a second resistor positioned between the first terminal node and a third terminal node. The resistance bridge also comprises a third resistor positioned between the second terminal node and a fourth terminal node. The resistance bridge also comprises a fourth resistor positioned between the third and fourth terminal nodes. The pressure sensor also comprises at least one current source associated with and electrically parallel to a respective one of the first through fourth resistors. The pressure sensor also comprises a switch associated with the at least one current source and configured to couple selectively the at least one current source to the respective one of the first through fourth resistors.

(6) In another aspect, a device is disclosed. The device comprises a pressure sensor. The pressure sensor comprises a resistance bridge. The resistance bridge comprises a first resistor positioned between a first terminal node and a second terminal node. The resistance bridge also comprises a second resistor positioned between the first terminal node and a third terminal node. The resistance bridge also comprises a third resistor positioned between the second terminal node and a fourth terminal node. The resistance bridge also comprises a fourth resistor positioned between the third and fourth terminal nodes. The pressure sensor also comprises at least one current source associated with and electrically parallel to a respective one of the first through fourth resistors. The pressure sensor also comprises a switch associated with the at least one current source and configured to couple selectively the at least one current source to the respective one of the first through fourth resistors. The device also comprises a temperature sensor. The device also comprises a control circuit configured to receive a signal from the temperature sensor and control the switch and the at least one current source to compensate for temperature variations based on the signal.

(7) In another aspect, a method for compensating for temperature variations in a pressure sensor is

disclosed. The method comprises detecting a temperature with a temperature sensor. The method also comprises selectively coupling a current source to a resistor in a resistance bridge based on the temperature.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is a circuit diagram of an exemplary conventional resistance bridge;
- (2) FIG. 1B is a graph of temperature versus offset voltage that shows how the offset voltage in the resistance bridge of FIG. 1A can vary with temperature;
- (3) FIG. 2 is block diagram of a device that includes an exemplary temperature-compensated resistance bridge according to the present disclosure;
- (4) FIG. 3 is a circuit diagram of a resistance bridge with temperature-compensating current sources according to exemplary aspects of the present disclosure; and
- (5) FIG. 4 is a graph of temperature versus offset voltage for the resistance bridge of FIG. 3.

DETAILED DESCRIPTION

(6) The embodiments set forth below represent the necessary information to enable those skilled in the art to practice the embodiments and illustrate the best mode of practicing the embodiments. Upon reading the following description in light of the accompanying drawing figures, those skilled in the art will understand the concepts of the disclosure and will recognize applications of these concepts not particularly addressed herein. It should be understood that these concepts and applications fall within the scope of the disclosure and the accompanying claims.

(7) It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present disclosure. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(8) It will be understood that when an element such as a layer, region, or substrate is referred to as being “on” or extending “onto” another element, it can be directly on or extend directly onto the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” or extending “directly onto” another element, there are no intervening elements present. Likewise, it will be understood that when an element such as a layer, region, or substrate is referred to as being “over” or extending “over” another element, it can be directly over or extend directly over the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly over” or extending “directly over” another element, there are no intervening elements present. It will also be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.

(9) Relative terms such as “below” or “above” or “upper” or “lower” or “horizontal” or “vertical” may be used herein to describe a relationship of one element, layer, or region to another element, layer, or region as illustrated in the Figures. It will be understood that these terms and those discussed above are intended to encompass different orientations of the device in addition to the orientation depicted in the Figures.

(10) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates

otherwise. It will be further understood that the terms “comprises,” “comprising,” “includes,” and/or “including” when used herein specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(11) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms used herein should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(12) Aspects disclosed in the detailed description include systems and methods for temperature coefficient of offset compensation for a resistance bridge. In particular, exemplary aspects of the present disclosure contemplate adding one or more current sources in parallel to resistance elements within a resistance bridge. The current source(s) may be selectively switched on and adjusted by a control circuit based on readings from a temperature sensor. In this fashion, the temperature induced variations in the resistance may be canceled or corrected allowing for better performance of the resistance bridge.

(13) Before addressing particular aspects of the present disclosure, an overview of some of the issues with a conventional system are provided with reference to FIGS. 1A and 1B. A discussion of exemplary aspects of the present disclosure begins below with reference to FIG. 2.

(14) In this regard, FIG. 1A is a circuit diagram of a circuit **100** that may, for example, be a Wheatstone bridge circuit and acts as a resistance bridge. Resistance bridges may act as pressure sensors in micro-electrical mechanical systems (MEMS). The circuit **100** may include a first resistor **102** having terminal nodes **104** and **106**. A second resistor **108** may couple to the terminal node **104**. A third resistor **110** may couple to the terminal node **106**. The circuit **100** may further include a fourth resistor **112** having terminal nodes **114** and **116**. The second resistor **108** may couple to the terminal node **114**, and the third resistor may couple to the terminal node **116**. The resistors **102**, **108**, **110**, and **112** may be formed from a piezoelectric material such that when pressure or force is applied to the circuit **100**, the resistances of the resistors **102**, **108**, **110**, and **112** may change.

(15) An excitation voltage (V_s) may be applied to the terminal nodes **104** and **116** and an offset voltage (V_{off}) may be generated between the terminal nodes **106** and **114**. When the resistors **102**, **108**, **110**, and **112** are balanced, V_{off} should be zero (0). When pressure is applied, the piezoelectric nature of the resistors **102**, **108**, **110**, and **112** causes the resistances to change (creating imbalance) and thus generate a value for V_{off} . Based on the value of V_{off} , the force being applied may be measured or approximated. In the abstract, this approach works well. However, the resistors **102**, **108**, **110**, and **112** may be temperature sensitive, and, as illustrated in FIG. 1B, this sensitivity may cause V_{off} to vary as a function of temperature even with no pressure applied.

(16) Specifically, FIG. 1B shows a graph **120** where V_{off} varies as a function of temperature. Specifically, given a temperature T_0 , V_{off} may be a first value **122**, but for a second temperature $T_1 > T_0$, V_{off} may be a second value **124** greater than the first value **122**. Such temperature-induced changes in V_{off} negatively impact the desired force measurements made by evaluation of V_{off} . That is, where there is both a force and temperature component to V_{off} , it may be difficult or impossible to isolate how much force is being applied to the circuit **100**.

(17) Exemplary aspects of the present disclosure add one or more current sources to a circuit to provide a compensation current through one or more of the resistances of a resistance bridge. By adjusting current through the resistors, balance may be restored to the bridge, allowing any measured V_{off} to be just a function of force, which in turn restores the functionality of the resistance bridge as a pressure sensor.

(18) In this regard, FIG. 2 is a block diagram of a device **200** having a current offset temperature

coefficient (TC) compensated pressure sensor **202** (more detail on this sensor **202** in FIG. 3, discussed below). A temperature sensor **204** may be present in the device **200** or located relatively close thereto and provide information related to temperature to a control circuit **206**. The control circuit **206** may provide an offset control signal to an offset control circuit **208**, which in turn controls elements within the sensor **202**. An output from the sensor **202** may be provided to an analog processing circuit **210** which provides a processed signal to the control circuit **206**.

(19) FIG. 3 illustrates the details of the sensor **202**. The sensor **202** may include a first resistor **302** having terminal nodes **304** and **306**. A second resistor **308** may couple to the terminal node **304**. A third resistor **310** may couple to the terminal node **306**. The sensor **202** may further include a fourth resistor **312** having terminal nodes **314** and **316**. The second resistor **308** may couple to the terminal node **314**, and the third resistor **310** may couple to the terminal node **316**. The resistors **302**, **308**, **310**, and **312** may be formed from a piezoelectric material such that when pressure or force is applied to the sensor **202**, the resistances of the resistors **302**, **308**, **310**, and **312** may change.

(20) One or more current sources **320(1)**-**320(4)** may be provided electrically in parallel with the resistors **302**, **308**, **310**, and **312**, respectively. Switches **322(1)**-**322(4)** may be provided to allow the current sources **320(1)**-**320(4)** to be activated selectively within the sensor **202**. The current sources **320(1)**-**320(4)** and the switches **322(1)**-**322(4)** may be controlled by the control circuit **206** or the offset control circuit **208**.

(21) In an exemplary aspect, the current sources **320(1)**-**320(4)** may be generated by a bandgap circuit, such that their temperature coefficient is well controlled and can be trimmed by using on-chip registers (not shown) or one time programmable (OTP) memory. It should further be appreciated that the current sources **320(1)**-**320(4)** have relatively large output impedances and thus have negligible effect on sensor sensitivity. Each current source **320(1)**-**320(4)** may be individually turned on or off through the associated switch **322(1)**-**322(4)**. Likewise, not all resistors **302**, **308**, **310**, **312** may have an associated current source **320(1)**-**320(4)**. While only current source **320(1)** is shown in solid lines with the other optional current sources **320(2)**-**320(4)** shown with dotted lines to imply optionality, it should be appreciated that any single one of the current sources **320(1)**-**320(4)** may be present with the other three omitted. Likewise, there may be two or three current sources present. However, for fine balancing, each resistor **302**, **308**, **310**, and **312** has its associated current source **320(1)**-**320(4)**. The current sources **320(1)**-**320(4)** may provide positive or negative output voltage offset to correct for temperature variation. In an exemplary aspect, current sources **320(1)**, **320(4)** are used for negative offset correction while current sources **320(2)**, **320(3)** are used for positive offset correction.

(22) It should be appreciated that the current sources **320(1)**-**320(4)** may be proportional-to-absolute temperature (PTAT) current sources or the current sources **320(1)**-**320(4)** may be conversely proportional-to-absolute temperature (CTAT) current sources. When a sensor **202** has a default positive temperature coefficient (i.e., without correction by the present disclosure), the PTAT current sources may be used. When a sensor **202** has a default negative temperature coefficient, CTAT current sources are used.

(23) While initial tests are not perfect, these tests show that using the current sources **320(1)**-**320(4)** provides a much more temperature insensitive sensor as shown by graph **400** in FIG. 4, where at temperature T_0 , $V_{off}(T)$ is approximately zero and likewise at T_1 , $V_{off}(T)$ is also approximately zero.

(24) Providing this temperature compensation effectively reduces or eliminates the dependence of V_{off} from temperature, thereby providing more linear and predictable operation. Such improved linearity improves the functionality of the sensor without requiring extensive external or over engineer precise resistors. Likewise, there is no need to integrate variability into a complementary metal oxide semiconductor (CMOS) device with its appurtenant margins for error.

(25) It is also noted that the operational steps described in any of the exemplary aspects herein are

described to provide examples and discussion. The operations described may be performed in numerous different sequences other than the illustrated sequences. Furthermore, operations described in a single operational step may actually be performed in a number of different steps. Additionally, one or more operational steps discussed in the exemplary aspects may be combined. It is to be understood that the operational steps illustrated in the flowchart diagrams may be subject to numerous different modifications as will be readily apparent to one of skill in the art. Those of skill in the art will also understand that information and signals may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

(26) The previous description of the disclosure is provided to enable any person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations. Thus, the disclosure is not intended to be limited to the examples and designs described herein but is to be accorded the widest scope consistent with the principles and novel features disclosed herein.

Claims

1. A pressure sensor, comprising: a resistance bridge comprising: a first resistor positioned between a first terminal node and a second terminal node; a second resistor positioned between the first terminal node and a third terminal node; a third resistor positioned between the second terminal node and a fourth terminal node; and a fourth resistor positioned between the third and fourth terminal nodes; at least one current source associated with and electrically parallel to a respective one of the first through fourth resistors; and a switch associated with the at least one current source and configured to couple selectively the at least one current source to the respective one of the first through fourth resistors.
2. The pressure sensor of claim 1, wherein the at least one current source comprises four current sources, each of the four current sources associated with and electrically parallel to respective ones of the first through fourth resistors.
3. The pressure sensor of claim 2, further comprising second through fourth switches associated with respective ones of the four current sources and configured to couple selectively the respective ones of the four current sources to respective ones of the first through fourth resistors.
4. The pressure sensor of claim 2, wherein a first of the four current sources comprises a proportional-to-absolute temperature (PTAT) current source.
5. The pressure sensor of claim 4, wherein the PTAT current source is configured to compensate an offset voltage based on a signal from a temperature sensor.
6. The pressure sensor of claim 2, wherein a first of the four current sources comprises a conversely proportional-to-absolute temperature (CTAT) current source.
7. The pressure sensor of claim 6, wherein a second of the four current sources comprises a proportional-to-absolute temperature (PTAT) current source.
8. The pressure sensor of claim 1, wherein the first through fourth resistors comprise piezoelectric material.
9. The pressure sensor of claim 1 integrated into a micro-electrical mechanical system (MEMS) device.
10. The pressure sensor of claim 1, further comprising a voltage source (Vs) coupled to the first and fourth terminal nodes.
11. A device comprising: a pressure sensor comprising: a resistance bridge comprising: a first resistor positioned between a first terminal node and a second terminal node; a second resistor positioned between the first terminal node and a third terminal node; a third resistor positioned

between the second terminal node and a fourth terminal node; and a fourth resistor positioned between the third and fourth terminal nodes; at least one current source associated with and electrically parallel to a respective one of the first through fourth resistors; and a switch associated with the at least one current source and configured to couple selectively the at least one current source to the respective one of the first through fourth resistors; a temperature sensor; and a control circuit configured to receive a signal from the temperature sensor and control the switch and the at least one current source to compensate for temperature variations based on the signal.

12. A method for compensating for temperature variations in a pressure sensor, the method comprising: detecting a temperature with a temperature sensor; and selectively coupling a current source to a resistor in a resistance bridge based on the temperature.

13. The method of claim 12, wherein selectively coupling the current source comprises activating a switch to couple the current source to the resistor.

14. The method of claim 12, wherein selectively coupling the current source comprises selectively coupling at least one of four current sources to a respective resistor in the resistance bridge.

15. The method of claim 12, wherein selectively coupling comprises selectively coupling a proportional-to-absolute temperature (PTAT) current source to the resistor.

16. The method of claim 12, wherein selectively coupling comprises selectively coupling a conversely proportional-to-absolute temperature (CTAT) current source to the resistor.
